

YASHIKA SINGH

102 1/2 Western Avenue, Floor 1, Jersey City, NJ 07307

+1(646)242-3188 • yashika6499@gmail.com • linkedin.com/in/yashika-singh-944700195 • yashika-portfolio.streamlit.app

Summary

Innovative and impact-driven AI/ML Engineer with 3+ years of experience building end-to-end AI solutions and data-driven products. Recognized for leading the design and deployment of an AI interviewer, delivering human-like, conversational AI experiences adopted by paying users. Uniquely skilled at bridging machine learning, generative AI, LLMs, and data science to build solutions.

Education

Yeshiva University, Katz School of Science and Health, New York, NY

Master of Science in Artificial Intelligence, GPA: 3.8/4.0

May 2025

Galgotias College of Engineering And Technology, Greater Noida, India

Bachelor of Technology in Computer Science, CGPA: 3.5/4.0

Aug 2021

Work Experience

Lead AI Engineer, Columbia University Summer Startup Program, New York

July 2025 - Aug 2025

- Collaborated with Columbia University student founders on a fintech startup initiative, providing AI leadership from concept to MVP.
- Lead the end-to-end development of Retrieval-Augmented Generation (RAG) pipelines with integrated LLMs, handling both backend architecture and frontend user experience.
- Delivered client-facing dashboards and technical documentation, and drove technical decision-making and roadmap planning for the MVP launch.

AI/ML Engineer Intern, Product Manager Accelerator, Boston, MA (Remote)

Jan 2025 - May 2025

- Designed and deployed AI-driven interview simulation workflows leveraging RAGs and LLMs (OpenAI, Claude) to generate context-aware product management and behavioral assessments. Scheduled for market release in July 2025, with paid customers onboarded.
- Applied NLP techniques to develop dynamic prompt engineering, follow-up logic, and structured candidate evaluation.
- Improved real-time response performance by 70% by optimizing model pipeline and collaborated with cross-functional teams to migrate business logic into production.
- Led research on multimodal AI interfaces, evaluating video avatar platforms and advanced TTS models to define the next cohort's roadmap for a more human-like, voice-and-visual interview experience.

Data Scientist, Axtria, Noida, India

Apr 2022 - Dec 2023

- Developed a K-Means clustering model using pharmaceutical engagement data to segment HCPs (Healthcare Professionals) based on behavioral patterns, enabling stakeholders to optimize marketing strategies, personalize outreach, and enhance overall HCP engagement.
- Developed and maintained ETL pipelines using Matillion, SQL and Python.
- Automated complex data processing workflows using DAGs in Apache Airflow, reducing manual intervention by 40% and enabled autonomous operations in dynamic environments.
- Integrated MLOps best practices with CI/CD pipelines to automate AI model training, deployment and monitoring on Databricks, ensuring scalability and reliability.

Data Analyst, Axtria, Noida, India

Aug 2021 - Apr 2022

- Optimized automated pipelines in Azure Data Factory and Apache Airflow, reducing data processing errors and improving overall efficiency.
- Supported in maintenance and advancement of the company's product Axtria Datamax, a cloud-based data management product.
- Created interactive dashboards and data visualizations using Power BI and Tableau to deliver insights and support decision-making for internal stakeholders.

Projects

Drowsiness Detection for Drivers: [GitHub Link]

Built a CNN-based, real-time system using TensorFlow to detect driver fatigue and improve road safety with validation accuracy around 95%.

Detection of Cardiomegaly in Dogs: [GitHub Link]

Created a deep learning model using custom CNN for X-ray analysis and automating veterinary diagnostics with an accuracy of 76.75%, surpassing the benchmark set by VGG16 (75%).

Skills and Certifications

Programming Languages: Python, C, C++, JavaScript, SQL

Developer Tools/Cloud: Apache Airflow (DAGs), Azure Data Factory, AWS, Google Cloud Platform (GCP), VS Code, Git, TensorFlow, Streamlit, Tableau, Power BI, PyTorch, Computer Vision, Anaconda, Jupyter, Google Colab, Pandas, Scikit-learn, Matplotlib, Databricks

Skills: Data Analysis, Machine Learning, Deep Learning, Natural Language Processing (NLP), Convolutional Neural Networks (CNN), Automatic Speech Recognition (ASR), Predictive Modeling, Exploratory Data Analysis (EDA), Whisper, Large Language Models (LLM), Prompt Engineering, Generative AI, Retrieval-Augmented Generation (RAG), MySQL, PostgreSQL, Extract-Transform-Load (ETL), HTML5, CSS